POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks: 100
Course: Object Oriented Software Engineering | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Suppose Pokhara University is planning to make a software application for managing programs, notices and extra curricular information. This application needs to be completed in a span of 6-9 months for the information sharing only and the rest of the features will be added in around 2 years. Propose a software development model for this software and justify your answer. 8

   b) What is software process framework? The framework activities are complemented by a number of umbrella activities. Explain each umbrella activities 7

2. a) Compute the function point value for a project with following information. 8

   Number of inputs: 450
   Number of outputs: 700
   Number of inquiries: 303
   Number of files: 210
   Number of external interfaces: 85

   Assume that all complexity adjustment values are average. If the productivity of the company is 85 FP/person-month and average salary is $620 per month then calculate the cost and effort required for the project. Assume necessary data if required.

   b) What are the processes involved while planning a project? Also explain the software feasibility with an appropriate example. 7

3. a) Obtain 1 level DFD for the following system of enchasing cheque in a bank. A customer presents a cheque to a clerk. The clerk checks the ledger containing all account numbers and makes sure whether the account number in the cheque is valid, whether adequate balance is there in the account to pay the cheque, and whether the signature is authentic. Having done these, the clerk gives the customer a token. The clerk also debits customer account by the amount specified on the cheque. If cash cannot be paid due to an error in the cheuqe, the cheque is returned. The token number is written on the top of the cheque and it is passed on the cashier. The cashier calls out token number, takes the customer signature, pays cash, enters cash paid in a ledger called daybook, and file the cheque. 9

   b) Differentiate between class diagram and object diagram with example 6

4. a) Describe the principle of Software Engineering and explain the the use of abstraction, modularity, functional independence in design concepts 8

   b) Describe the structure based modeling explaining its components with an appropriate example. 7

5. a) What is the ISO standard? Explain the goals of software quality assurance 8

   b) What do you understand about software reliability? Compare between verification and validation testing. 7

6. a) What do you mean by test case and test suite? Prepare a test case for a simple login form. 8

   b) Explain the emerging trends in software Engineering technology evolution. 7

7. Write short notes on: **(Any two)** 2×5
   a) Design concepts
   b) Use-case diagram
   c) SCM

Available on notes.arpanadhikari7.com.np